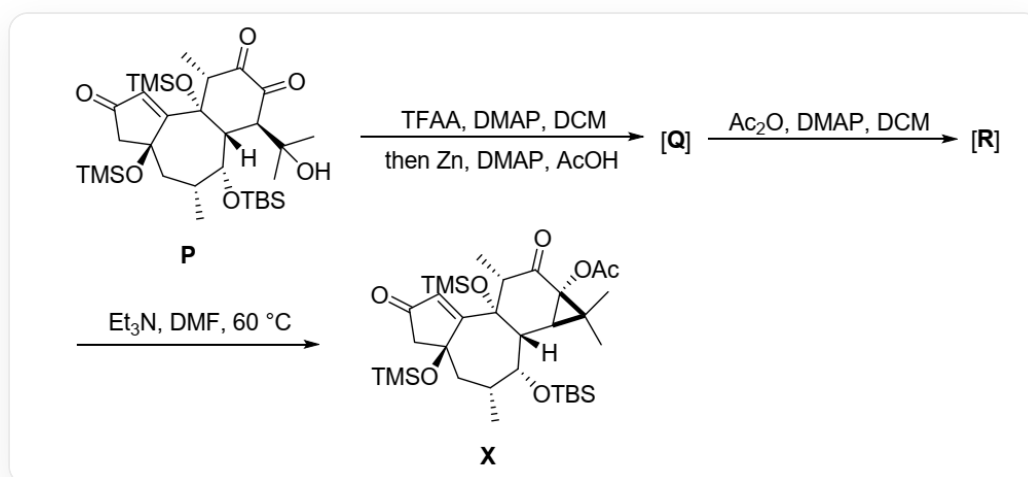


Question

Compound **P** first reacts with TFAA, followed by the action of *Zn* to generate **Q**. **Q** generates **R** with a ketal structure under the action of acetic anhydride. **R** obtains the final product **X** with a fused hexacyclic-tricyclic system under the action of triethylamine. It is known that a new five-membered ring is produced in both the reaction from **P** to **Q** and from **Q** to **R**.



C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)O)C(C([C@H]3C)=O)=O)[C@@H]2O[Si](C)(C)C(C)(C)C)O[Si](C)(C)C)=CC(C1)=O)C first reacts with trifluoroacetic anhydride and DMAP in DCM, followed by reaction with zinc, DMAP, and acetic acid to obtain **Q**. **Q** reacts with acetic anhydride and DMAP in DCM to obtain **R**. **R** and triethylamine react in DMF at 60 degrees Celsius to obtain **X**: C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)4C)[C@]4(C([C@H]3C)=O)OC(C)=O)[C@@H]2O[Si](C)(C)C(C)(C)C)O[Si](C)(C)C)=CC(C1)=O)C

The following statements are made:

1. The role of *Zn* in the step of generating **Q** is as a reducing agent.
2. There is a trans-vicinal diol structure in **Q**.
3. There is a carbon atom bonded to three oxygen atoms in **R**.
4. It is known that the process of generating **X** successively undergoes two negatively charged intermediates **Y1** and **Y2**, then **Y2** has 3 rings.

Select the option containing all correct statements.

A. All statements are incorrect.

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **N**

Detailed Explanation

TFAA is a strong acylation reagent. Under DMAP catalysis, the tertiary alcohol hydroxyl group in **P** is more nucleophilic than the oxygen of the enol ether or the enol oxygen of the 1,3-diketone, and will preferentially react with TFAA to generate a trifluoroacetate. Subsequently, zinc reduction of the vicinal diketone in acetic acid produces an enol, which attacks the trifluoroacetate to generate a five-membered ring.

CHECKPOINT

1 PTS

Zn acts as a reducing agent to reduce the vicinal diketone, statement 1 is correct

During the nucleophilic attack, zinc ion chelation controls the oxygen of the trifluoroacetate and the enol oxygen produced by the vicinal diketone to be on the same side, resulting in **Q**: `C[Si](C)(O[C@@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)O[C@@]4(C(F)(F)F)O)[C@]4(C([C@H]3C)=O)O)[C@@H]2O[Si](C)(C)C(C)(C)O[Si](C)(C)C)=CC(C1)=O)C`. It has a *cis*-vicinal diol structure.

CHECKPOINT

1 PTS

The structure of **Q** is `C[Si](C)(O[C@@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)O[C@@]4(C(F)(F)F)O)[C@]4(C([C@H]3C)=O)O)[C@@H]2O[Si](C)(C)C(C)(C)O[Si](C)(C)C)=CC(C1)=O)C`. It has a *cis*-vicinal diol structure. Statement 2 is incorrect

Under the action of acetic anhydride, the newly generated hydroxyl group of **Q** is converted to an acetate. Due to the vicinal diol structure, the other hydroxyl group can attack the ester group, generating a five-membered ring, resulting in **R**: `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H]4[C@]5(C([C@H]3C)=O)O[C@](O)(C)O[C@]5(C(F)(F)F)OC(C)4C)[C@@H]2O[Si](C)(C)C(C)C)O[Si](C)(C)C)=CC(C1)=O)C`, in which there is a carbon atom bonded to three oxygen atoms.

CHECKPOINT

1 PTS

The structure of **R** is `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H]4[C@]5(C([C@H]3C)=O)O[C@](O)(C)O[C@]5(C(F)(F)F)OC(C)4C)[C@@H]2O[Si](C)(C)C(C)C)O[Si](C)(C)C)=CC(C1)=O)C`, in which there is a carbon atom bonded to three oxygen atoms. Statement 3 is correct

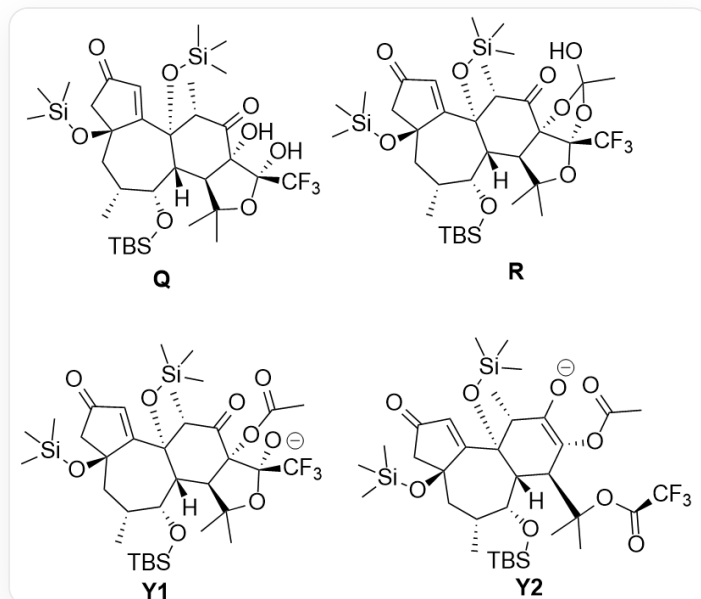
Under the action of base, the hydroxyl hydrogen on the orthoester is abstracted, and the orthoester decomposes to generate intermediate **Y1**: `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)O[C@@]4(C(F)(F)F)[O-])[C@]4(C([C@H]3C)=O)OC(C)=O)[C@@H]2O[Si](C)(C)C(C)(C)C)O[Si](C)(C)C)=CC(C1)=O)C`. Subsequently, the negative charge on the oxygen attacks the carbon-carbon bond, and a retro-Aldol reaction occurs, opening the five-membered ring to generate enol **Y2**: `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)OC(C(F)(F)F)=O)C(OC(C)=O)=C([C@H]3C)[O-])[C@@H]2O[Si](C)(C)C(C)(C)C)O[Si](C)(C)C)=CC(C1)=O)C`, which has 3 rings. The enolate anion attacks the carbon atom connected to the trifluoroacetate oxygen, generating a three-membered ring, and trifluoroacetate is eliminated, generating **X**.

CHECKPOINT

1 PTS

The structure of **Y2** is `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)OC(C(F)(F)F)=O)C(OC(C)=O)=C([C@H]3C)[O-])[C@@H]2O[Si](C)(C)C(C)(C)C)O[Si](C)(C)C)=CC(C1)=O)C`, which has 3 rings. Statement 4 is correct

Statements 1, 3, and 4 are correct, choose N.



Q : `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)O[C@@]4(C(F)(F)F)O)[C@]4(C([C@H]3C)=O)O)[C@@H]2O[Si](C)(C)C(C)(C)O[Si](C)(C)C=CC(C1)=O)C` ; **R** : `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H]4[C@]5(C([C@H]3C)=O)O[C@](O)(C)O[C@]5(C(F)(F)F)OC(C)4C)[C@@H]2O[Si](C)(C)C(C)(C)O[Si](C)(C)C=CC(C1)=O)C` ; **Y1** : `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)O[C@@]4(C(F)(F)F)[O-])[C@]4(C([C@H]3C)=O)OC(C)=O)[C@@H]2O[Si](C)(C)C(C)(C)O[Si](C)(C)C=CC(C1)=O)C` ; **Y2** : `C[Si](C)(O[C@]1(C[C@H]2C)C([C@@]3([C@]([H])([C@H](C(C)(C)OC(C(F)(F)F)=O)C(OC(C)=O)=C([C@H]3C)[O-])[C@@H]2O[Si](C)(C)C(C)(C)O[Si](C)(C)C=CC(C1)=O)C`